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Smooth Lagrangian crack band model with softening spress–sprain relation and crack width prediction[☆]

Houlin Xu [a,1,2], Yang Zhao [a,1], Anh Tay Nguyen [b], Zdeněk P. Bažant [c,*]

[a] Department of Civil and Environmental Engineering, Northwestern University, 2145 Sheridan Road, Evanston, 60208, IL, USA
[b] Department of Mechanical Engineering, Northwestern University, 2145 Sheridan Road, Evanston, 60208, IL, USA
[c] Departments of Civil and Environmental Engineering, Mechanical Engineering, and Materials Science and Engineering, Northwestern University, 2145 Sheridan Road, CEE/A135, Evanston, 60208, IL, USA

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ABSTRACT

Fracture development in solids is characterized by a finite fracture process zone (FPZ) ahead of the crack tip, which progressively localizes into a discontinuous crack surface behind the FPZ. The classical crack band model (CBM) controls the strain localization by the minimum allowed finite element size. However, it suffers from a certain degree of mesh bias in constitutive parameter calibration and crack path prediction. To overcome these limitations, the smooth Lagrangian crack band model (slCBM) has been proposed. It regularizes the size of the FPZ by introducing a much finer finite element subdivision at and near the FPZ and an additional sprain energy that penalizes excessive curvature of the displacement field. The original slCBM employs a linear spress–sprain relation, but its regularization length determined by the maximum curvature increases under large deformation, producing artificial postpeak resistance. This also hinders the formation of localized cracks, limiting the applicability to problems such as hydraulic fracturing. To address these issues, the present work introduces an updated slCBM with postpeak softening damage in which the sprain stiffness decays exponentially with either the volumetric strain ϵ_v or the sprain magnitude $|\xi|$. This enhancement enables consistent FPZ regularization ahead of the crack tip while allowing, after sufficient damage, complete strain localization within a row of sub-elements, thereby achieving a smooth transition from distributed damage to discrete crack formation. Numerical demonstrations confirm the model's ability to reproduce smooth strain evolution in partially damaged regions and localized crack formation in fully damaged zones. The proposed framework allows a realistic and consistent prediction of the crack width, which controls the rate of water flow along hydraulic fracture.

1. Introduction

In the framework of Linear Elastic Fracture Mechanics (LEFM) [1], the material outside a sharp crack tip is assumed to behave elastically, and the crack tip fields are characterized by singular stresses and strains that decay with the inverse square root of the distance to the crack tip [2]. This idealization has proved sufficient for brittle materials under small-scale yielding conditions and at negligible crack-parallel stresses [3,4].

The challenge, however, lies in accurate modeling of the complex process of damage initiation and progression, especially in quasi-brittle materials that exhibit pronounced microcracking, inelastic deformation, and softening as elastic deformation transits to complete rupture [5]. In such materials, fracture is characterized by a finite fracture process zone (FPZ) led by interacting defects (microcracks, pores, aggregates, fibers, etc.), and the effective behavior is strongly influenced by internal length scales associated with large crack-parallel stresses and with the heterogeneity of the microstructure [6,7] which causes the energy dissipation to be spread out over a fracture process zone (FPZ) of a finite width [6,8]. The wide FPZ, with its sensitivity to the crack-parallel stress, cannot be captured by the conventional LEFM-based approaches (including the phase-field and cohesive crack models) [9–12], resulting in inconsistent fracture toughness measurements.

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* Corresponding author.
E-mail address: z-bazant@northwestern.edu (Z.P. Bažant).

[1] These authors contributed equally to this work.

[2] Current address: Hopkins Extreme Materials Institute, Malone Hall 140, Johns Hopkins University, 3400 N Charles St, Baltimore, MD 21218

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Comparisons of the crack band, cohesive, linear elastic, phase-field and peridynamic models showed the crack band model to be superior in reproducing eleven most characteristic fracture experiments [13,14]. However, the subsequent gap test [3,4] was not reproduced well by any of these models. This has revealed that the distribution of damage across the crack band was important.

As a remedy, the smooth Lagrangian crack band model (slCBM) [15, 16] introduced two new concepts: First, instead of fixing the width of damage zone, the damage localization was limited by the resistance to increasing curvature (or the Hessian) of the displacement vector field, characterized by $u_{i,jk}$ (u_i continuum displacement vector, i, j, k = cartesian coordinate subscripts). Second, while $u_{i,jk}$ is a third order derivative (same as in the strain gradient model for plastic-hardening metals [17–19]), the slCBM uses only finite elements with shape functions of C_0 continuity because, for fracture modeling, the lowest order continuity is generally the best (which is why the isogeometric analysis (IGA), with cubic spline shape functions [20], is not very good for fracture modeling).

The idea is to model both the displacement vector and its gradient by linear shape functions with C_0 continuity. These functions can, of course, be matched only in the average sense. This match is rendered possible by using the Lagrangian multiplier [15]. The actual gradient as a linear function calculated from the nodal displacements is matched to the actual linear gradient by means of the Lagrange multiplier that must be generalized as a second-order tensor. The idea of matching of the gradients was originally suggested by equating the mean of the gradient to the finite difference expression expressing the gradient in terms of the nodal displacements, and then enforcing such equality variationally. The advantage of this approach is that the Lagrange multiplier behaves like a constant external force (similar to gravity), thus contributing a zero energy density (unlike the alternative of a penalty function).

In some important problems, e.g., in the hydraulic fracturing of rock, aka fracking, or in assessing the corrosion due to ingress of corrosive fluids, one needs to predict the width of the final crack after the stresses in the damage zone get reduced to zero. In the classical crack band model, the crack width was estimated as the total displacement across the crack band minus the displacement due to the elastic strains which were uniform across the band [6,21–23]. This method, however, can have a significant error, especially if nonlinear behavior is considered; e.g., an error by a factor 2 is possible, which would lead to an 8-fold error in the laminar flow rate along the crack [24].

Here we eliminate such errors in crack width by taking into account the postpeak softening. We attribute this softening either to an increase of the volumetric strain (as in [25]) or to an increase of the sprain, and compare both alternatives.

2. Summary of the smooth Lagrangian Crack Band Model (slCBM)

The recently developed constitutive model for sprain must reproduce the resistance to curvature of the displacement field. Specifically, this is the resistance to the curvature tensor ξ_{ijk}, which represents the second gradient (or Hessian) of the displacement vector field $\boldsymbol{u}$ (rather than the gradient of the strain field as in the strain gradient models [17,19,26]). The resisting force-like variable, which is work-conjugate to the sprain tensor, s_{ijk}, has been called [16] the "*spress*" (borrowed from the root of Italian word "e*spresso*").

Briefly, the theory decomposes the Helmholtz continuum energy density into two terms

$$\bar{\Psi}(\epsilon, \xi) = \Psi(\epsilon) + \Phi(\xi) \tag{1}$$

where $\Psi(\epsilon)$ represents the standard strain energy density, and $\Phi(\xi)$ represents the sprain energy density; ϵ is the strain tensor; $\boldsymbol{\xi}$ is the sprain tensor defined as $\xi_{ijk} = l_0 u_{i,jk}$, and l_0 is a regularization length. Specifically,

$$\Phi(\xi) = \int s_{ijk} \mathrm{d}\xi_{ijk}, \qquad s_{ijk} = \kappa \left\langle |\xi| - C \right\rangle \frac{\xi_{ijk}}{|\xi|} \tag{2}$$

In this equation, κ is the sprain stiffness. C is a spress threshold. The spress is activated when $|\xi| > C$.

For prepeak loading, a linear spress–sprain relation was shown in previous studies to be adequate. But it is not for postpeak loading. It would cause the growth of an unbounded sprain and an excessive expansion of the nonlocalized damage zone, especially at large strain. Ultimately, it would prevent a localized crack with two surfaces from forming behind the crack tip, which is essential for modeling the fluid flow through hydraulic cracks.

Despite the implied third displacement gradients, we prefer to avoid high-order element methods. For fracture modeling, linear shape functions with C_0 continuity are generally optimal, offering greater flexibility in capturing displacement discontinuities. Therefore, as introduced in [15,16], we consider the displacements as well as their approximate gradients ζ_{ij} to be represented by linear interpolation functions weakly constrained by a Lagrange multiplier, λ_{ij}, which is an additional independent variable - a second-order tensor (this multiplier was logically introduced as variational restatement of equality of the finite difference expression for the displacement gradient to the mean approximate gradient).

In terms of internal and external (Helmholtz free) energy densities,

$$W_{\text{in}} = \int_V \Psi[\epsilon(\boldsymbol{u})] \, \mathrm{d}V + \int_V \Phi[\xi(\boldsymbol{\zeta})] \, \mathrm{d}V - \int_V \lambda : (\nabla \boldsymbol{u} - \boldsymbol{\zeta}) \, \mathrm{d}V \tag{3}$$

$$W_{\text{ex}} = \int_V f(\boldsymbol{x}) \cdot \boldsymbol{u} \, \mathrm{d}V \tag{4}$$

From these equations it is clear that the Lagrangian multiplier λ must here be conceived as a second-order tensor.

The variational formulation of the Helmholtz free energy functional with spress–sprain relation leads to a system of three coupled work-balance equations, derived from the principle of virtual work. By taking the first variations of the Helmholtz free energy with respect to $\boldsymbol{u}$, $\boldsymbol{\zeta}$ and λ, we obtain the following variational equations [16]:

$$\begin{aligned} &\int_V \left\{ \sigma_{ij} \cdot \delta\epsilon_{ij} - f_i \cdot \delta u_i - \lambda_{ij} \cdot \delta u_{i,j} \right\} \mathrm{d}V + \int_S p_i u_i \mathrm{d}S \\ &\quad = \int_V \left\{ -\sigma_{ij,j} - f_i + \lambda_{ij,j} \right\} \cdot \delta u_i \, \mathrm{d}V + \int_S (\dots) \, \mathrm{d}S = 0 \end{aligned} \tag{5}$$

$$\begin{aligned} &\int_V \left\{ s_{ijk} \cdot \delta\xi_{ijk} + \lambda_{ij} \cdot \delta\zeta_{ij} \right\} \mathrm{d}V \\ &\quad = \int_V \left\{ -l_0 s_{ijk,k} + \lambda_{ij} \right\} \cdot \delta\zeta_{ij} \, \mathrm{d}V + \int_S (\dots) \, \mathrm{d}S = 0 \end{aligned} \tag{6}$$

$$\int_V - \left\{ u_{i,j} - \zeta_{ij} \right\} \cdot \delta\lambda_{ij} \, \mathrm{d}V = 0 \tag{7}$$

These three equations represent the local balance of momentum, the constraint of the sprain tensor, and the weak constraint of the displacement gradient field to the independent gradient-approximation field, respectively. The variational principle ensures that the system achieves equilibrium while maintaining the geometric restrictions imposed by the spress–sprain relation.

The slCBM with sprain energy eliminates the need for identifying the adjacent nodes used in the original formulation [25] and the independent field ζ_{ij}, which approximates the actual strain gradient $u_{i,j}$. The fields ζ_{ij} and $u_{i,j}$ are weakly constrained by a Lagrange multiplier tensor λ_{ij}, while the gradient $\zeta_{ij,k}$ represents the curvature (or second gradient, the Hessian) of the displacement vector u_i. This formulation not only obviates the necessity of applying curvature-resisting sprain forces on adjacent nodes but also allows both u_i and ζ_{ij} fields to be assigned the lowest inter-element continuity, C_0, which is generally advantageous for fracture modeling.

It should be noted that the slCBM, adopting a linear spress–sprain relation with a constant sprain stiffness κ_0, is suitable only for simulating fracture development up to the maximum external load $P_{\max}$. As also observed in phase-field formulations and various gradient-based regularization schemes [27–32], "linear penalization" mechanisms may eventually constrain the material response throughout the fracture

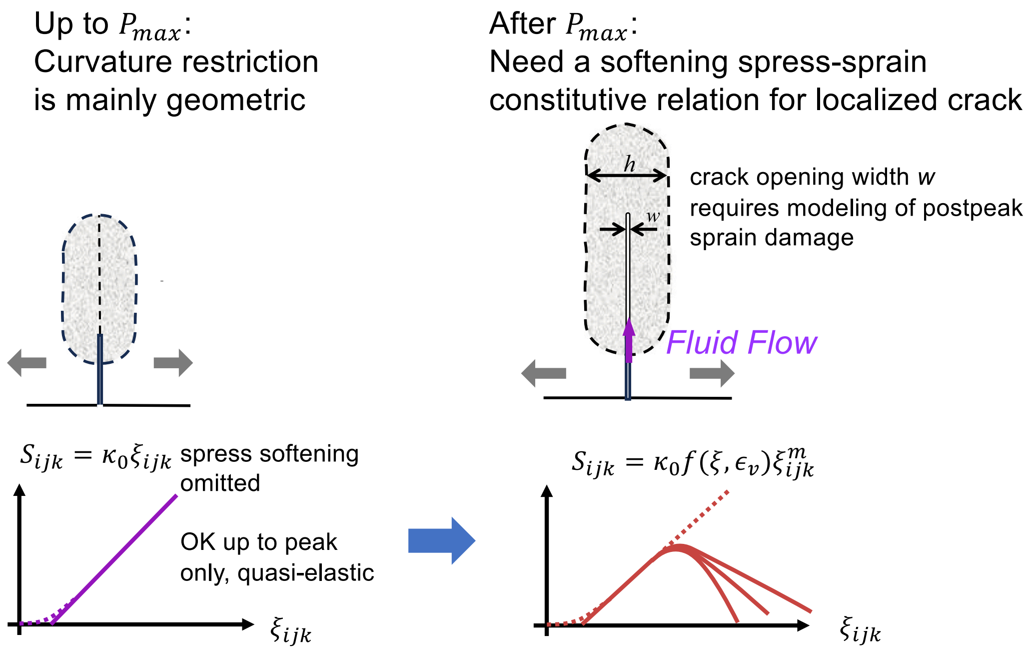


Fig. 1. Requirements of spress–sprain relation for fracture process simulation: A constant sprain stiffness is acceptable up to peak load; the softening of spress is needed to capture postpeak crack localization and crack width estimation.

process. To capture the postpeak response and to allow a localized fracture to form behind the advancing crack tip, a postpeak softening of spress is required.

A comparison between the two cases of postpeak softening is shown in Fig. 1. Achieving proper crack localization, and therefore an accurate estimation of crack opening, is essential for predicting permeability evolution in rocks during fracturing, which is vital for applications such as hydraulic fracturing in unconventional reservoirs [33–35].

A natural extension of the present model is to allow the sprain stiffness to gradually decrease with either the sprain magnitude $|\xi|$ or the volumetric strain ϵ_v. This reduction progressively relaxes the nonlocal regularization based on structural curvature control during the postpeak stage, enabling the damage zone to contract and eventually allowing a fully localized crack of finite width to emerge.

3. Extended sprain theory with exponentially decaying sprain stiffness

As introduced in previous sections, the linear spress–sprain relation in slCBM possesses limitations causing the regularization size to expand at large sprain. Thus the model does not allow the formation of localized cracks in the fully damaged region behind the crack tip. To address these problems, here we propose an extension of the slCBM by considering the softening of sprain stiffness, driven by either the sprain magnitude $|\xi|$ or the volumetric strain ϵ_v.

The sprain stiffness can be extended to decay with the sprain magnitude, and a simple candidate is to introduce an exponential term,

$$s = \kappa e^{-A\langle|\xi|-C\rangle} \langle|\xi| - C\rangle \frac{\xi}{|\xi|} \tag{8}$$

The exponential term ensures that the sprain stiffness decays to zero at large sprain, which can capture the softening of the spress when the fracture develops. Equivalently, one can also define nonlinear sprain energy function Φ that has an asymptote when $|\xi| \to \infty$, which guarantees that the spress would soften to zero at large sprain.

It has been proposed in [25] for the original smooth crack band model that the sprain stiffness κ can also degrade with the normal strain. The reasoning behind it is that when fracture develops, the material in the fracture process zone will be fully damaged with large tensile normal or volumetric strain, at which time there should be no curvature resistance.

Linking the sprain stiffness to the strain measure is an implicit way to reflect the degradation of sprain stiffness with material damage. In this work, we propose an exponential decay of the sprain stiffness driven by the expansive volumetric strain. We introduce the spress–sprain relation as

$$s = \kappa e^{-A\langle\epsilon_v\rangle} \langle|\xi| - C\rangle \frac{\xi}{|\xi|} \tag{9}$$

The non-negative parameter A quantifies the significance of the degradation mechanism's influence associated with the current $|\xi|$ or ϵ_v . This formula ensures that the spress s_{ijk} degrades to zero when the material is fully damaged and large volumetric strain ϵ_v has occurred. Note, however, that involving the strain components in the spress–sprain relation would also lead to a reciprocal sprain component in the stress–strain relation derived from the free energy. This may make the formulation too complicated. So we consider only a one-directional effect of strain on spress.

Remark on the selection of softening schemes: The selection between the two proposed softening schemes depends on the specific focus of the modeling effort and the characteristics of the material and loading conditions:

1. The ϵ_v-driven softening scheme captures the influence of volumetric strain on the postpeak response. It is particularly suitable for volume-sensitive materials, where dilatation plays a significant role in the fracture process, such as porous materials (e.g., concrete, rocks, foams) or materials exhibiting pressure-dependent failure behavior.
2. The ξ-driven softening scheme captures the influence of curvature (second gradient of displacement) on the softening behavior. It is more appropriate for loading scenarios where volumetric strain remains relatively stable, such as torsion, shear-dominated fracture, or Mode II/III cracking, where the curvature

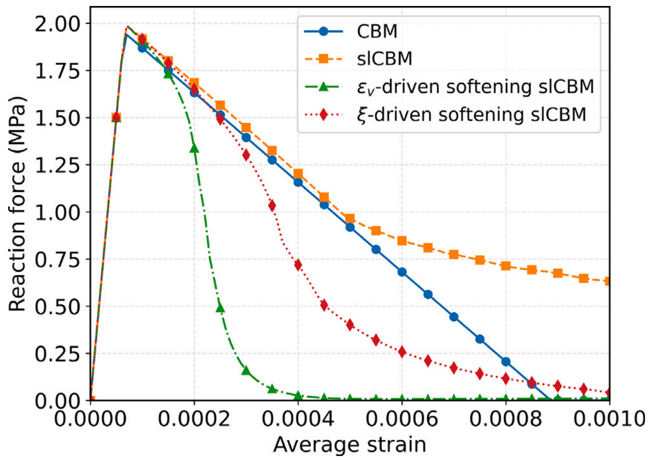


Fig. 2. Evolution of reaction stress with applied external strain to the bar.

of the displacement field governs the localization process. This scheme may be better suited for materials like fine-grained ceramics or certain metals under shear loading.

4. One-dimensional bar extension

Let us now consider the example of one dimensional extension of a bar with length $l = 0.1$ m using different fracture models. The material properties of the bar are: Young's modulus $E = 30$ GPa, tensile strength $f_t = 2$ MPa, and the postpeak response softens linearly, decaying to zero stress at $\epsilon_f = 4.47 \times 10^{-3}$. For slCBM methods with either constant or softening sprain stiffness, the bar is decomposed into 101 finite elements with one weak element placed in the middle with a lower tensile strength $f_t' = 0.975 f_t$ to trigger strain localization. For comparison, the CBM model is constructed using 5 elements, since the corresponding CBM element size matches the intrinsic regularization length adopted in the slCBM models. The bar is extended monotonically to a maximum average strain $\epsilon_{\max} = 1 \times 10^{-3}$, and four formulations are examined: the conventional CBM method, the slCBM with constant sprain stiffness, the softening slCBM with sprain stiffness degraded by sprain magnitude, and the softening slCBM with sprain stiffness degraded by volumetric strain. The parameters adopted for the regularization using the sprain energy theory are summarized as follows: $l_0 = 0.01$ m, $\kappa = 95$ kPa, $C = 0$. Regarding the parameter controlling the exponential decay rate of the sprain stiffness, $A = 830$ for ϵ_v-driven softening slCBM, whereas $A = 83$ for ξ-driven softening slCBM. Note that the numerical values of A in the two softening mechanisms are not directly comparable. This is because the ϵ_v-driven mechanism is governed by the first derivative of displacement (strain), while the ξ-driven mechanism involves the second derivative of displacement (sprain, related to curvature). During material damage, the characteristic magnitudes of these two quantities differ substantially, which naturally requires different values of A to achieve comparable softening effects.

The force–displacement curves for the four cases are shown in Fig. 2. By comparing the results predicted with the CBM and the slCBM methods, one may note that the peak force achieved by slCBM is slightly higher, which is close to the tensile strength f_t, whereas that for CBM is close to the strength of the middle element f_t'. This is due to the fact that the sprain energy spreads out the damage zone to multiple elements within the regularization zone. The postpeak responses for slCBM and CBM are in agreement except for larger strain, where the regularization zone size starts to increase. For cases with softening of the spress–sprain relations, either driven by sprain or volumetric strain, exponential softening prevents the expansion of the regularization zone and the reaction force will drop to zero at large applied strain. Compared to CBM, these two cases also have less dissipated energy interpreted as the area under the curves compared to CBM, as the localization band for these two cases is narrower than the element size of CBM.

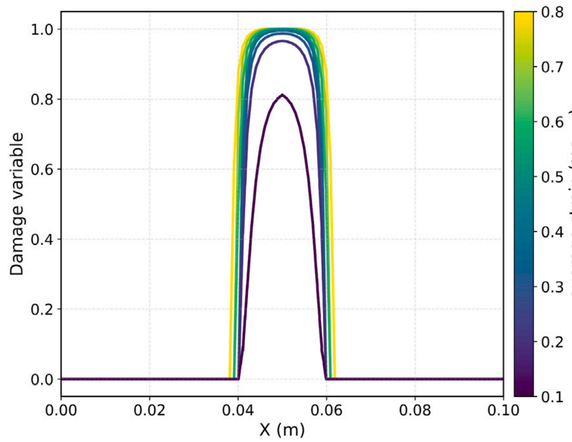
Fig. 3 exhibits the evolution of the damage variable and axial strain profiles for the three different cases where the fracture process zone is regularized using the sprain energy. For the slCBM model with a constant sprain stiffness, it transpires that the damage first grows within the span of the regularization length, and the strain profile inside appears to be a bell shaped curve spanning the same width. With the increase of applied extension, the regularization length also increases, which induces the postpeak response deviating from CBM observed in Fig. 2. For the case where a softening spress–sprain relation driven by ξ is adopted, the regularization length shrinks when the sprain stiffness decreases, and the ultimate damage zone with localized strain is about half of the initial regularization length. Lastly, for the case where the sprain stiffness degrades with the volumetric strain, the damage zone as well as the strain can localize into three elements, which corresponds to the minimum number of elements that can reflect the smooth Lagrangian structure to represent strain and sprain.

Fig. 4 shows the evolution of sprain and spress profiles for the three different cases where the fracture process zone is regularized with the sprain theory. For the conventional slCBM, two antisymmetric bell-shaped profiles form around the weak element for both the sprain and spress. For the case with a ξ-driven softening spress–sprain relation, the sprain tends to localize around the maximum points of the bell shape, and the spress tends to decrease at these points due to the softening mechanism. This is the reason why the damage zone and strain localization band are controlled within nearly half of the initial regularization length. Finally, for the case where the sprain stiffness degrades with the volumetric strain, the sprain stiffness softens the most at the weak element in the middle since the maximum volumetric strain always occurs at the weak element. The degradation of the sprain stiffness changes the distribution of the sprain profile from a bell shape to be more skewed and concentrated toward the weak element, and the spress near the weak element decreases and allows a narrow strain localization.

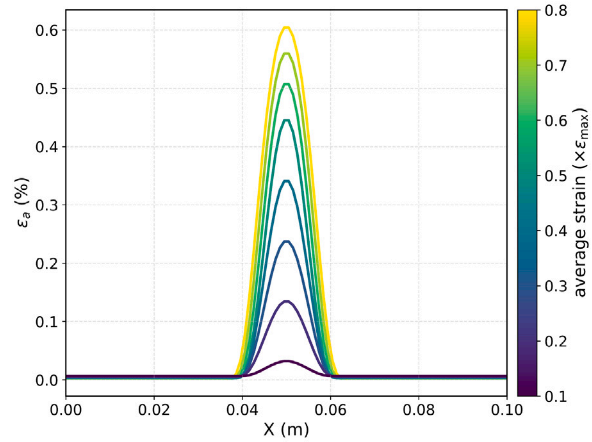
5. 2D three-point-bend test simulation

2D simulations of three-point-bend tests on a pre-notched specimen are conducted to compare the performance of the slCBM formulations with different ways of spress softening. The specimen used in the simulation is shown in Fig. 5, which is a rectangular domain with height $D = 0.2$ m and span $s = 0.5$ m. A straight notch is cut from the midpoint of the bottom edge, extending vertically upward into the specimen to a length of $a = 0.06$ m, with a uniform width of $w = 6.7$ mm. The material properties used for the simulation domain are as follows: Young's modulus $E = 37$ GPa, Poisson's ratio $\nu = 0.21$, tensile strength $f_t = 3.9$ MPa, and mode-I fracture energy $G_f = 85$ J/m^2. In terms of the parameters adopted for the spress–sprain relations, we have $\kappa = 0.6$ MPa, $C = 0$. For the softening parameters, we have $A = 20$ for the sprain-driven softening case and $A = 200$ for the volumetric strain-driven softening case, respectively. The regularization length scale $l_0 = 20$ mm, which is 3-times the notch size of the three-point-bend specimen.

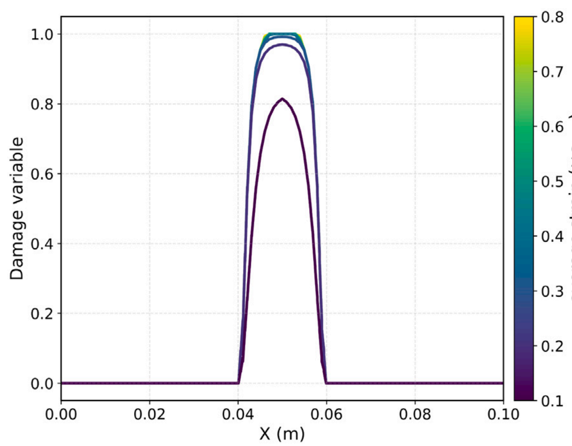
The simulated force–displacement curves obtained from different slCBM formulations are summarized in Fig. 6. The results show that all three cases exhibit very similar pre-peak behavior, although the conventional slCBM produces a slightly higher peak load. The differences become much more pronounced in the postpeak stage. In the conventional slCBM with constant sprain stiffness, the spress increases without bound as the sprain grows, which enlarges the effective regularization size and prevents a localized crack surface from forming behind the crack tip. This leads to an unrealistic residual structural



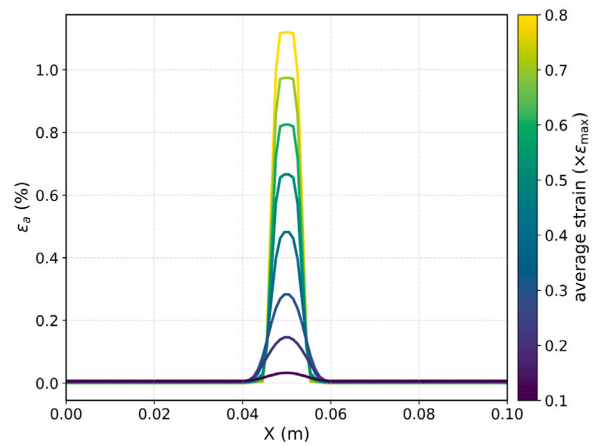
(a) slCBM: Damage



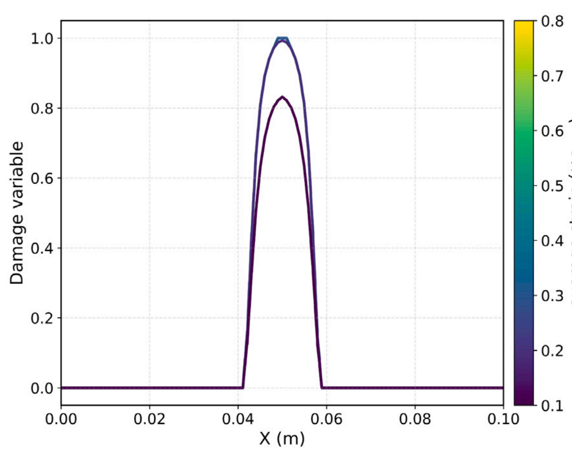
(b) slCBM: Axial strain



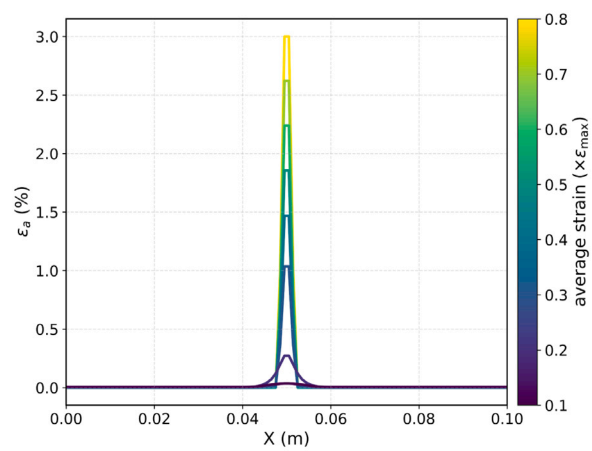
(c) ξ-driven softening slCBM: Damage



(d) ξ-driven softening slCBM: Axial strain

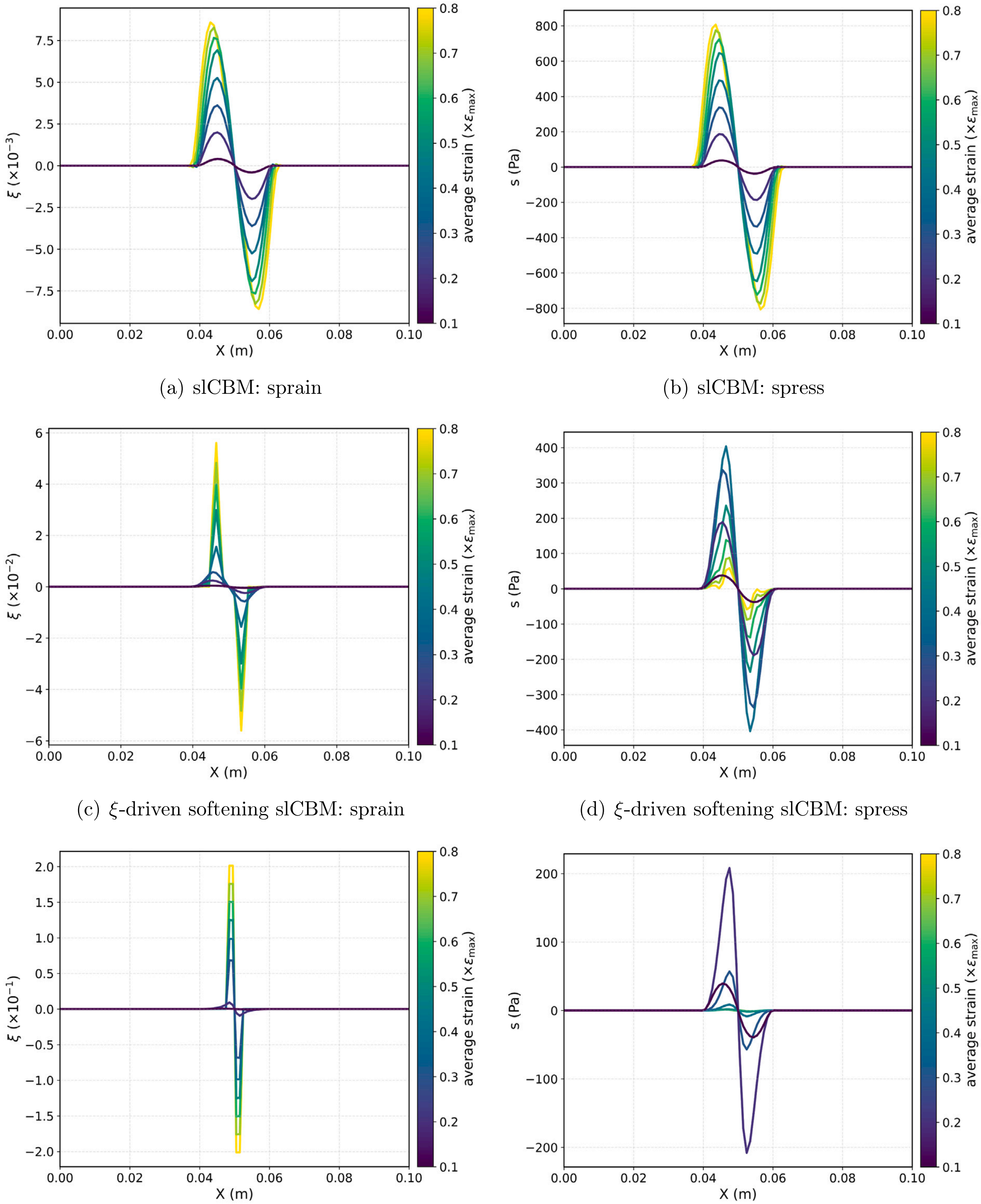


(e) ϵ_v-driven softening slCBM: Damage



(f) ϵ_v-driven softening slCBM: Axial strain

Fig. 3. Comparison of the evolution of damage variable and axial strain profiles. Left: Damage variable; Right: Axial strain. Top: slCBM; Middle: ξ-driven softening slCBM; Bottom: ϵ_v-driven softening slCBM.



(a) slCBM: sprain

(b) slCBM: spress

(c) ξ-driven softening slCBM: sprain

(d) ξ-driven softening slCBM: spress

(e) ϵ_v-driven softening slCBM: sprain

(f) ϵ_v-driven softening slCBM: spress

Fig. 4. Comparison of the evolution of sprain and spress profiles. Left: sprain; Right: spress. Top: slCBM; Middle: ξ-driven softening slCBM; Bottom: ϵ_v-driven softening slCBM.

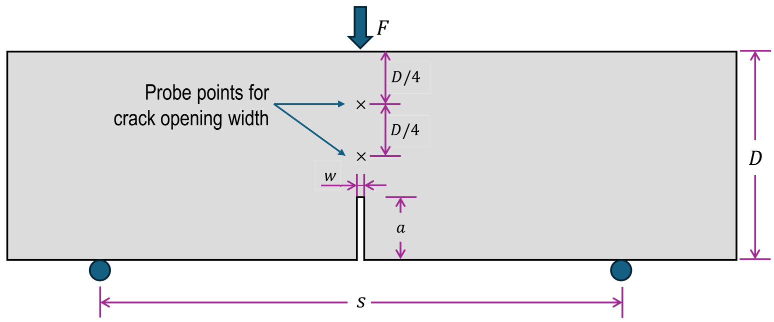


Fig. 5. Illustration of the three-point-bend test.

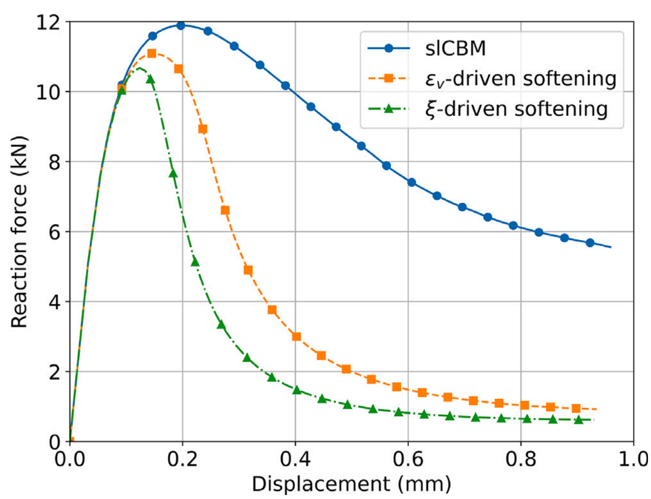


Fig. 6. Force–displacement curves from three point bending simulations using the slCBM with different spress softening strategies.

resistance. In contrast, the slCBM formulations with a softening spress–sprain law, whether driven by sprain or volumetric strain, allow the regularization to diminish and enable proper crack localization. As a result, they produce a more realistic post peak response, as illustrated in Fig. 6.

Fig. 7 shows the evolution of the strain component ϵ_{yy} along the crack opening y-direction, plotted as filled contours, together with the spress component s_{yyy} that controls the crack band width, shown as contour lines. For the conventional slCBM with a constant sprain stiffness, Fig. 7(a) indicates that the strain field retains a candle light shape and the damage zone continues to broaden, preventing the formation of a fully localized crack. In contrast, for the two softening slCBM formulations, Fig. 7(b,c) shows that once the crack begins to grow, spress degradation occurs primarily at and immediately behind the crack tip, which allows the strain to collapse into a narrow localized band. Both softening approaches produce clear crack localization, but with different widths: when the spress softening is driven by volumetric strain, the crack collapses into a single column of elements, whereas when the spress softening is driven by sprain magnitude, the localization band retains approximately the same width as the initial notch. This difference arises from the distinct spatial distributions of the softening variable in the two formulations, which constrain the width of the damage zone in different ways.

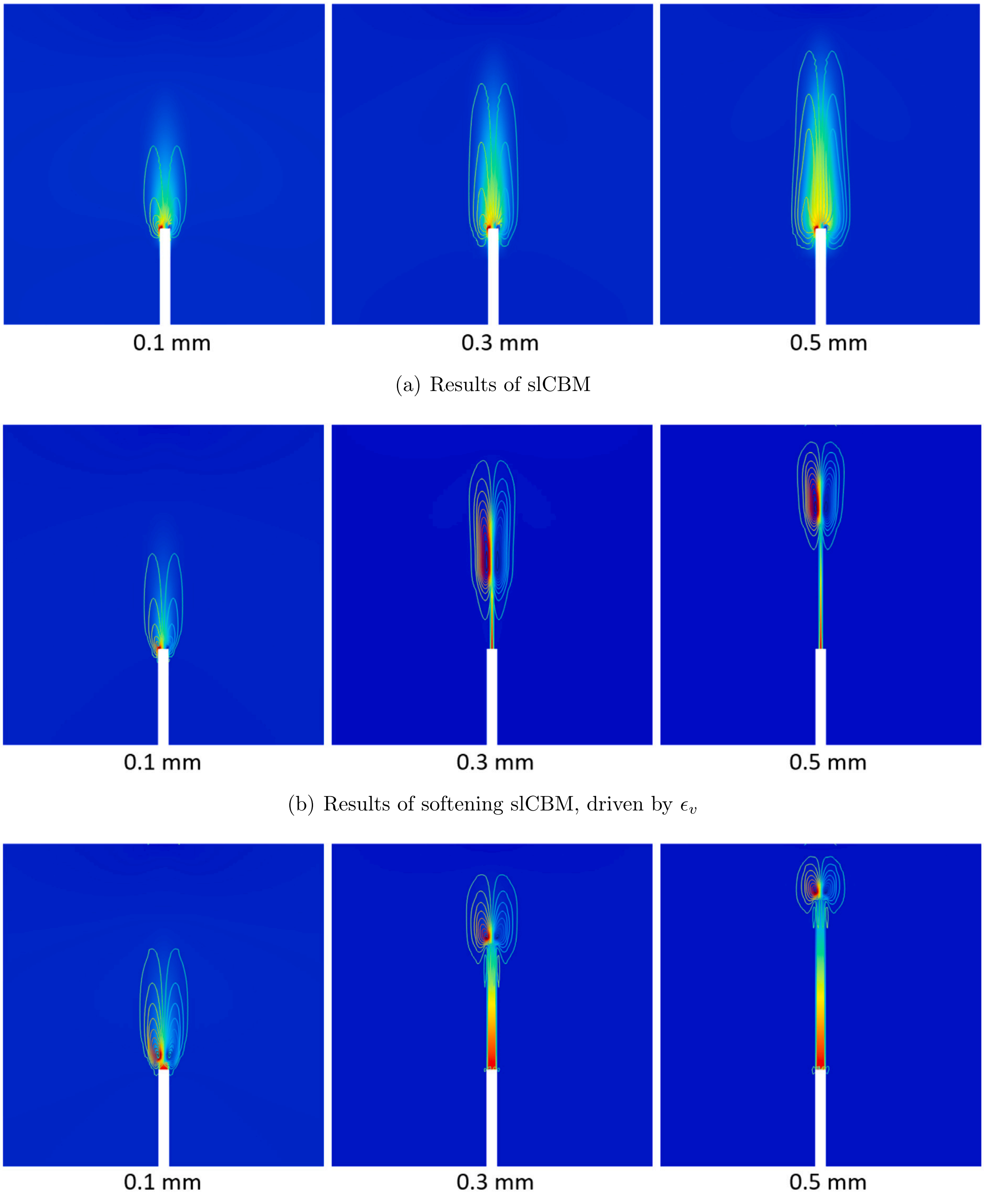
Convergence check: Fracture fundamentally arises from material softening in the constitutive behavior. This causes the governing equations of solid mechanics to lose ellipticity and shift toward hyperbolicity. Regularization methods that aim to achieve mesh independence, such as the slCBM, phase-field models, strain gradient formulations, and other nonlocal approaches, seek to preserve mathematical well-posedness by maintaining the elliptic nature of the governing partial differential equations. In the proposed framework, mesh dependence is moderated through the sprain energy, which controls strain localization at the onset of softening. As spress continues to degrade and a distinct crack begins to form, however, the regularizing effect weakens and mesh sensitivity can reemerge. This motivates a careful examination of the mesh sensitivity associated with the proposed softening slCBM schemes.

Fig. 8 exhibits the finite element meshes used for the mesh sensitivity check, zoomed in at the vicinity of the notch tip. The crack path ahead of the notch tip has been decomposed into 3, 5 or 7 columns of sub-elements, respectively. Three-point bending simulations using the slCBM with the two proposed spress softening schemes were performed on each of these meshes.

Note that, for practical applications, the choice of mesh size should consider the specific application context. While finer meshes improve crack width resolution during damage evolution, coupled problems (e.g., hydraulic fracturing) may require the mesh size to be compatible with the characteristic length scale of fluid flow. Therefore, a universally recommended ratio of mesh size to regularization length can be difficult to provide.

Fig. 9 shows the force–displacement curves recorded in the simulations. Despite consistent pre-peak response, differences in peak values and postpeak slopes become evident, which is expected to occur, due to the diminishing effect of the regularization (from spress) at the location of complete material fracture. Although the simulations use identical material and model parameters, the curves begin to diverge after a certain point in the softening regime. Finer discretization of the notch leads to a steeper postpeak response, as the effective crack band area decreases with the sub-element size, resulting in lower energy dissipation.

Fig. 10 presents the evolution of crack opening width with an increasing compressive displacement at two probe points along the crack path, located at $\frac{1}{2}D$ and $\frac{3}{4}D$ along the pre-notch. The crack opening width is computed by integrating ϵ_{yy}, the strain component in the crack opening direction, across the regularization zone. The results show that when material damage is still developing at the probe point, the simulated crack opening width exhibits mesh dependence. Once the material becomes fully damaged, however, the crack opening width recovers the mesh independent behavior.



(a) Results of slCBM

(b) Results of softening slCBM, driven by ϵ_v

(c) Results of softening slCBM, driven by ξ

Fig. 7. Evolution of strain component ϵ_{yy} (continuous contours) and the spress component s_{yyy} (discrete contours) evaluated at the applied compression of 0.1, 0.3, 0.5 mm. y−direction refers to crack opening direction.

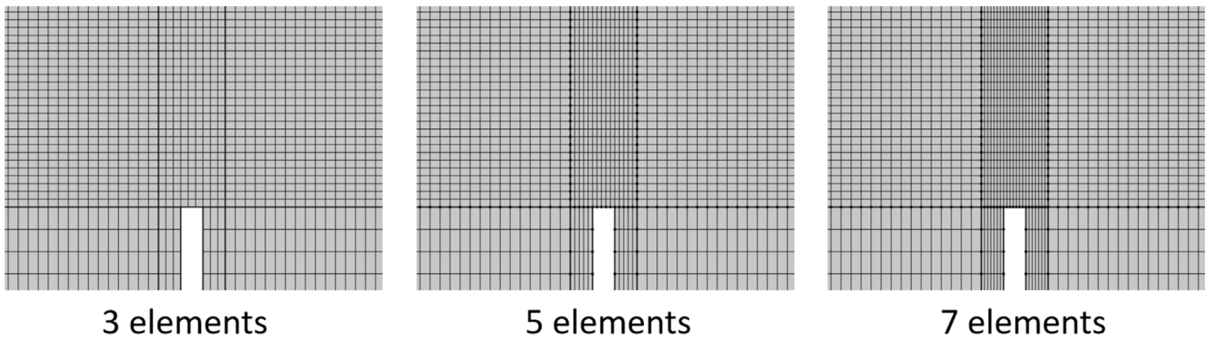


Fig. 8. Enlarged views of the mesh in the vicinity of the notch tip for the three meshes used in the sensitivity analysis. The potential crack path ahead of the notch tip is discretized into 3, 5, and 7 columns of subelements, respectively.

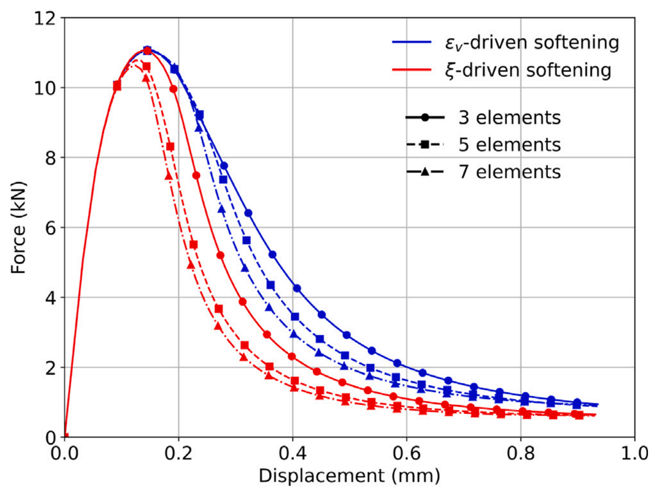


Fig. 9. Force–displacement curves in three-point-bend tests using a different mesh size and spress softening type.

6. 2D four-point-bend Iosipescu shear test simulation

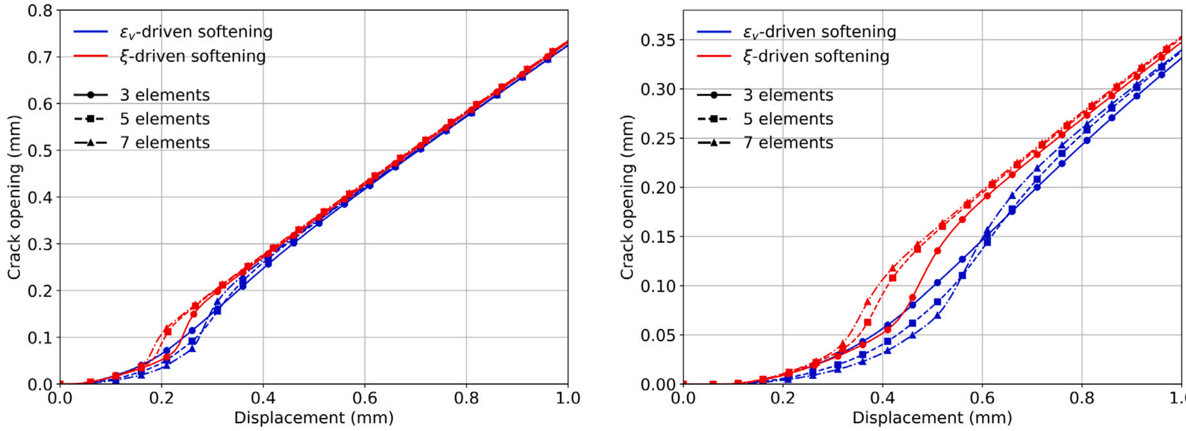
Another example that can be investigated is the four-point-bend (4PB) Iosipescu shear test. The specimen used in the simulation is shown in Fig. 11, which shows a rectangular domain with height D = 0.2 m and span s = 0.50 m. Two opposing straight notches are cut from the midpoints of the top and bottom edges, extending vertically toward the center of the specimen, each with a length of a = 0.06 m and a uniform width of w = 6.7 mm. A pair of antisymmetric loads F are applied near the notches, both anchored at a distance of d = 0.025 m from the symmetry axis of the structure and distributed over loading plates. The material properties used for the simulation domain are as follows: Young's modulus E = 37 GPa, Poisson's ratio ν = 0.21, tensile strength f_t = 3.9 MPa, and mode II fracture energy G_f = 400 J/m^2. In terms of the parameters used for the spress–sprain relation, we have κ = 0.02 MPa. The regularization length scale l_0 = 20 mm.

The sprain-driven (or ξ-driven) spress–sprain softening law is adopted in this study for two reasons: (1) the volumetric strain remains relatively stable under the present loading condition, and (2) the localization process is governed by the curvature of the displacement field. In Fig. 12, a parametric study of the parameter A is conducted with A = 2.2, 1.8, and 1.0 to illustrate the effect of the softening of the sprain stiffness. It can be observed that a larger value of A results in a steeper post-peak response, as the sprain stiffness degrades more rapidly with increasing sprain magnitude. Conversely, a smaller value of A leads to more gradual post-peak softening behavior. In the asymptotic limit, as A approaches zero, the model behavior converges toward the linear spress–sprain relation of the conventional slCBM (as shown in Fig. 4 of [16]).

The crack profile results also demonstrate that the fracture ultimately localizes into a single column of sub-elements within the central ligament (for an appropriate value of A). This confirms that the enhanced formulation with softening spress–sprain relation successfully achieves both critical objectives: retaining the beneficial strain-smoothing effects during intermediate damage evolution phases, while permitting complete, visible, strain localization at ultimate failure.

7. Conclusions

1. Fracture propagation in solids involves different length scales. The fracture process zone (FPZ) ahead of the crack tip has a finite width, whereas cracks far behind the crack tip must localize into a distinct crack bordered by two surfaces (in 3D) or lines (in 2D).
2. The crack band model (CBM) regularizes the fracture development with the element size as the characteristic length scale, which suffers from mesh bias. The smooth Lagrangian crack band model (slCBM) addresses this problem by using a regularization with displacement curvature control.
3. It is recommended to use the slCBM in the prepeak regime, as it yields a continuously increasing curvature resistance, even after the fracture is fully developed, with large strain and sprain. Such a formulation works well up to almost the peak load but suppresses the formation of a localized crack behind the advancing crack tip. This limitation indicates the need for a nonlinear, softening spress–sprain relation.
4. The proposed modification is to adopt a nonlinear spress–sprain relation with an exponentially decaying sprain stiffness, driven either by the volumetric strain magnitude or by the sprain magnitude. This allows the damage band to eventually localize into single stack of subdivided finite elements at the end of the postpeak structural response.
5. In this way one achieves both critical objectives for realistic fracture modeling: The beneficial strain-smoothing effects of spress during the intermediate damage evolution phases is retained, with a complete strain localization at the complete ultimate failure. This combination enhances the robustness and accuracy, important for practical applications, particularly when complex quasi-brittle fracture is involved.



(a) Probe point at $1/2D$ (b) Probe point at $3/4D$

Fig. 10. Evolution of crack opening width given different mesh size and spress softening type, evaluated at two probe points along the crack path: (a) Probe point at $\frac{1}{2}D$ along the pre-notch; (b) Probe point at $\frac{3}{4}D$ along the pre-notch.

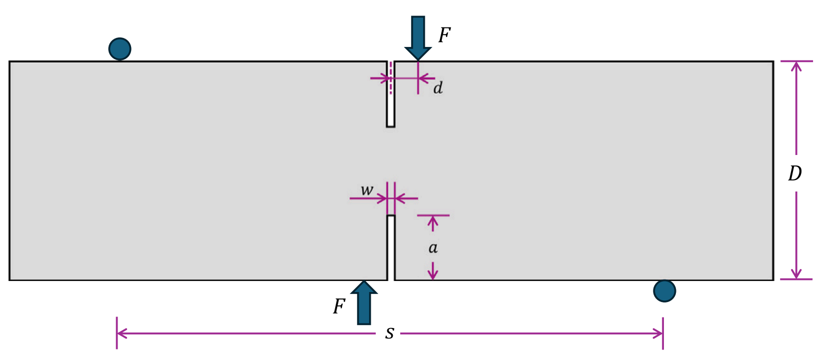


Fig. 11. Illustration of the Mode II four-point-bend (4PB) Iosipescu shear test.

6. The mesh sensitivity analysis shows that the proposed method remains mesh-size independent during the prepeak loading and produces nearly the same peak load across different mesh sizes.
7. The size of the subdivided finite elements must be smaller than the expected width of the final crack. If not, there will be local sensitivity to the chosen size of the subdivided finite elements.
8. Estimation of the crack opening width using the proposed method can be mesh dependent during the damage evolution at a given point along the crack path. However, once the material at that point reaches complete damage, the crack width converges and becomes effectively independent of the mesh.

CRediT authorship contribution statement

Houlin Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yang Zhao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anh Tay Nguyen:** Writing – review & editing, Validation, Investigation, Data curation, Conceptualization. **Zdeněk P. Bažant:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

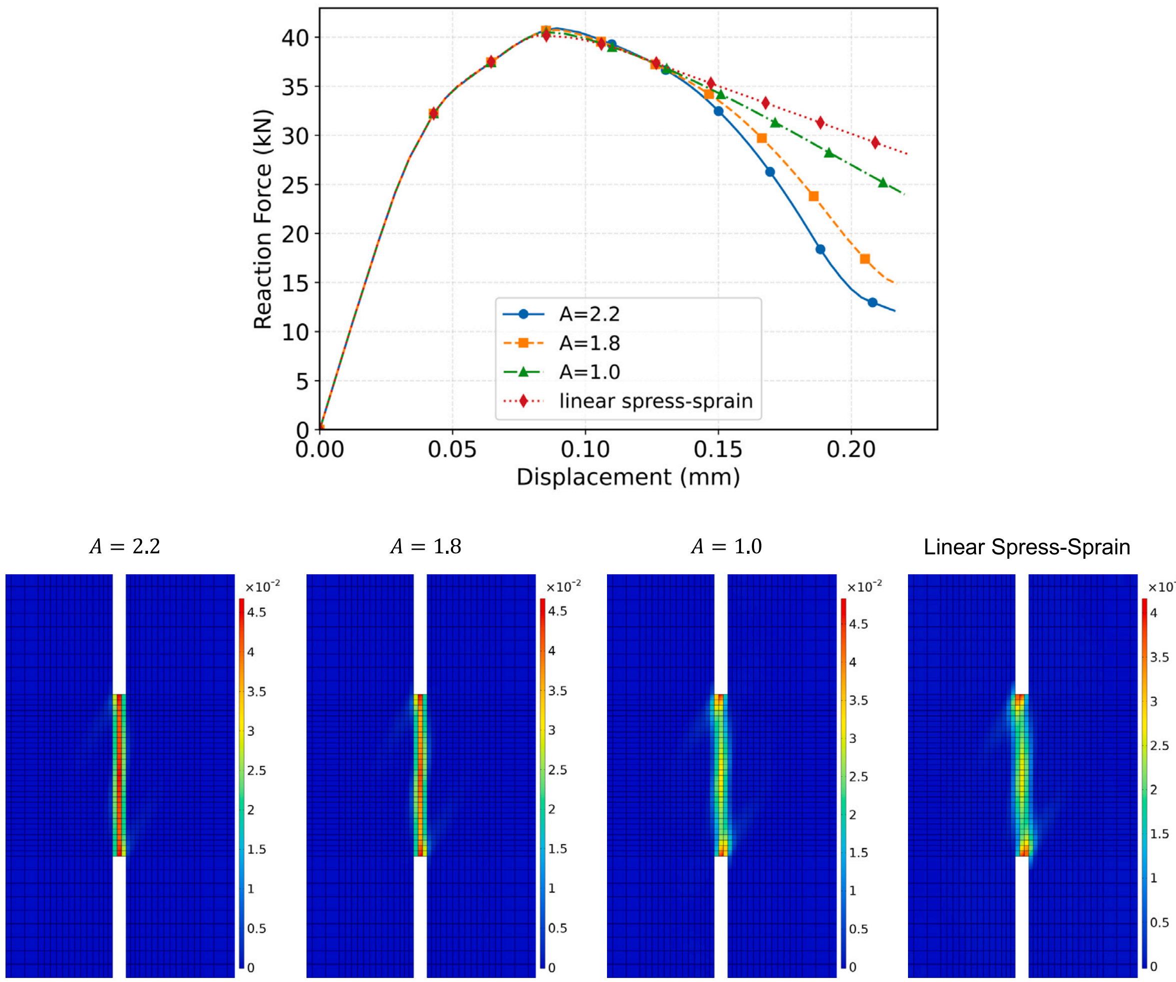

Fig. 12. Parametric study on the softening parameter A under shear loading. Top: Effect of different values of parameter A on the force vs. displacement curve; Bottom: Effect of different values of parameter A on the final crack formation. When A is very small, the result is asymptotic to that of the linear spress–sprain relation.

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